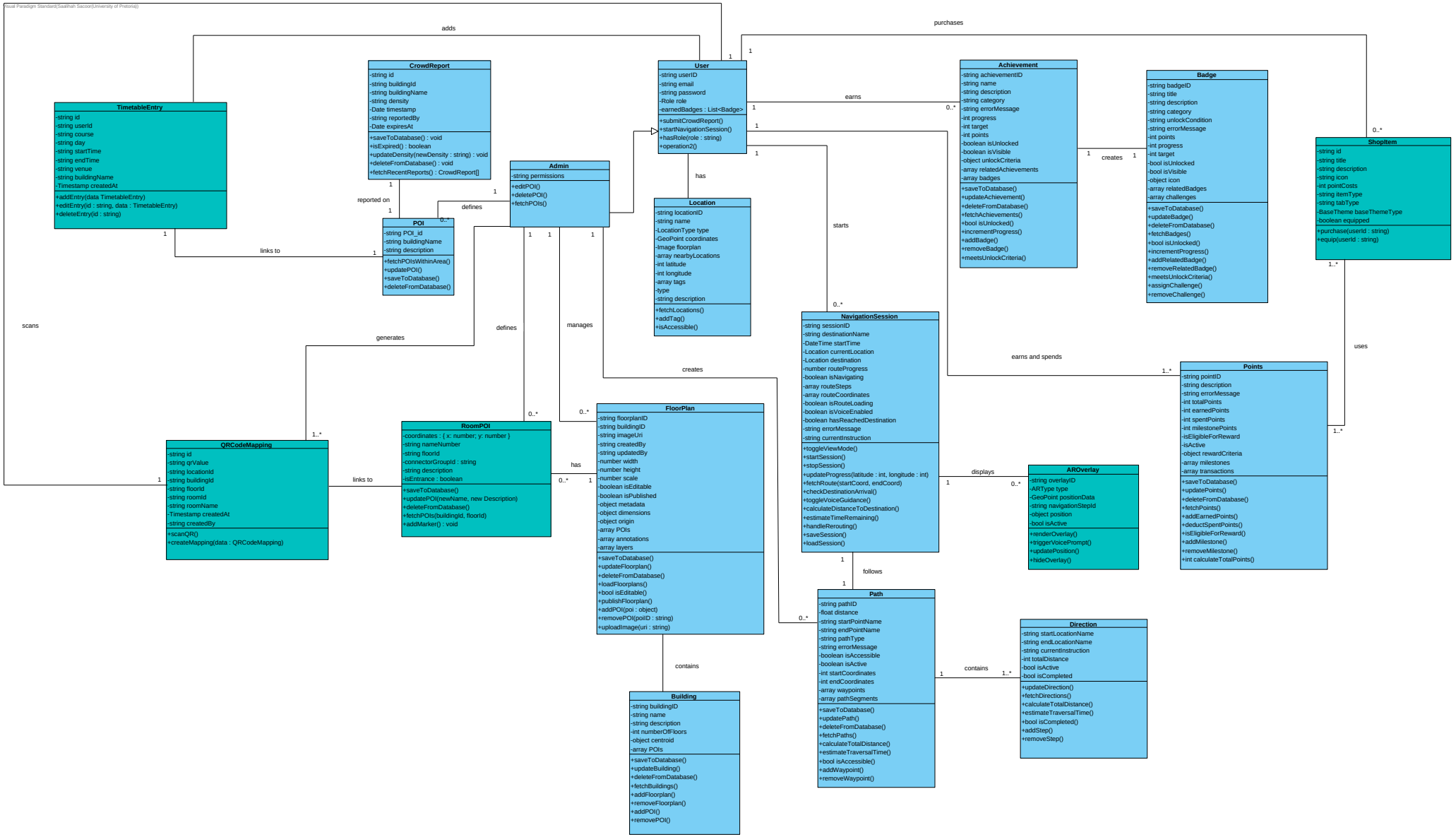
Visual Paradigm Standard(Saalihah Sacoor(University of Pretoria))
adds
purchases
CrowdReport
-string id
-string buildingId
-string buildingName
-string density
-Date timestamp
-string reportedBy
-Date expiresAt
+saveToDatabase() : void
+isExpired() : boolean
+updateDensity(newDensity : string) : void
+deleteFromDatabase() : void
+fetchRecentReports() : CrowdReport[]
User
-string userID
-string email
-string password
-Role role
-earnedBadges : List<Badge>
+submitCrowdReport()
+startNavigationSession()
+hasRole(role : string)
+operation2()
Achievement
-string achievementID
-string name
-string description
-string category
-string errorMessage
-int progress
-int target
-int points
-boolean isUnlocked
-boolean isVisible
-object unlockCriteria
-array relatedAchievements
-array badges
+saveToDatabase()
+updateAchievement()
+deleteFromDatabase()
+fetchAchievements()
+bool isUnlocked()
+incrementProgress()
+addBadge()
+removeBadge()
+meetsUnlockCriteria()
Badge
-string badgeID
-string title
-string description
-string category
-string unlockCondition
-string errorMessage
-int points
-int progress
-int target
-bool isUnlocked
-bool isVisible
-object icon
-array relatedBadges
-array challenges
+saveToDatabase()
+updateBadge()
+deleteFromDatabase()
+fetchBadges()
+bool isUnlocked()
+incrementProgress()
+addRelatedBadge()
+removeRelatedBadge()
+meetsUnlockCriteria()
+assignChallenge()
+removeChallenge()
ShopItem
-string id
-string title
-string description
-string icon
-int pointCosts
-string itemType
-string tabType
-BaseTheme baseThemeType
-boolean equipped
+purchase(userId : string)
+equip(userId : string)
TimetableEntry
-string id
-string userId
-string course
-string day
-string startTime
-string endTime
-string venue
-string buildingName
-Timestamp createdAt
+addEntry(data TimetableEntry)
+editEntry(id : string, data : TimetableEntry)
+deleteEntry(id : string)
Admin
-string permissions
+editPOI()
+deletePOI()
+fetchPOIs()
Location
-string locationID
-string name
-LocationType type
-GeoPoint coordinates
-Image floorplan
-array nearbyLocations
-int latitude
-int longitude
-array tags
-type
-string description
+fetchLocations()
+addTag()
+isAccessible()
POI
-string POI_id
-string buildingName
-string description
+fetchPOIsWithinArea()
+updatePOI()
+saveToDatabase()
+deleteFromDatabase()
NavigationSession
-string sessionID
-string destinationName
-DateTime startTime
-Location currentLocation
-Location destination
-number routeProgress
-boolean isNavigating
-array routeSteps
-array routeCoordinates
-boolean isRouteLoading
-boolean isVoiceEnabled
-boolean hasReachedDestination
-string errorMessage
-string currentInstruction
+toggleViewMode()
+startSession()
+stopSession()
+updateProgress(latitude : int, longitude : int)
+fetchRoute(startCoord, endCoord)
+checkDestinationArrival()
+toggleVoiceGuidance()
+calculateDistanceToDestination()
+estimateTimeRemaining()
+handleRerouting()
+saveSession()
+loadSession()
Points
-string pointID
-string description
-string errorMessage
-int totalPoints
-int earnedPoints
-int spentPoints
-int milestonePoints
-isEligibleForReward
-isActive
-object rewardCriteria
-array milestones
-array transactions
+saveToDatabase()
+updatePoints()
+deleteFromDatabase()
+fetchPoints()
+addEarnedPoints()
+deductSpentPoints()
+isEligibleForReward()
+addMilestone()
+removeMilestone()
+int calculateTotalPoints()
QRCodeMapping
-string id
-string qrValue
-string locationId
-string buildingId
-string floorId
-string roomId
-string roomName
-Timestamp createdAt
-string createdBy
+scanQR()
+createMapping(data : QRCodeMapping)
RoomPOI
-coordinates : { x: number; y: number }
-string nameNumber
-string floorId
-connectorGroupId : string
-string description
-isEntrance : boolean
+saveToDatabase()
+updatePOI(newName, new Description)
+deleteFromDatabase()
+fetchPOIs(buildingId, floorId)
+addMarker() : void
FloorPlan
-string floorplanID
-string buildingID
-string imageUri
-string createdBy
-string updatedBy
-number width
-number height
-number scale
-boolean isEditable
-boolean isPublished
-object metadata
-object dimensions
-object origin
-array POIs
-array annotations
-array layers
+saveToDatabase()
+updateFloorplan()
+deleteFromDatabase()
+loadFloorplans()
+bool isEditable()
+publishFloorplan()
+addPOI(poi : object)
+removePOI(poiID : string)
+uploadImage(uri : string)
AROverlay
-string overlayID
-ARType type
-GeoPoint positionData
-string navigationStepId
-object position
-bool isActive
+renderOverlay()
+triggerVoicePrompt()
+updatePosition()
+hideOverlay()
Path
-string pathID
-float distance
-string startPointName
-string endPointName
-string pathType
-string errorMessage
-boolean isAccessible
-boolean isActive
-int startCoordinates
-int endCoordinates
-array waypoints
-array pathSegments
+saveToDatabase()
+updatePath()
+deleteFromDatabase()
+fetchPaths()
+calculateTotalDistance()
+estimateTraversalTime()
+bool isAccessible()
+addWaypoint()
+removeWaypoint()
Direction
-string startLocationName
-string endLocationName
-string currentInstruction
-int totalDistance
-bool isActive
-bool isCompleted
+updateDirection()
+fetchDirections()
+calculateTotalDistance()
+estimateTraversalTime()
+bool isCompleted()
+addStep()
+removeStep()
Building
-string buildingID
-string name
-string description
-int numberOfFloors
-object centroid
-array POIs
+saveToDatabase()
+updateBuilding()
+deleteFromDatabase()
+fetchBuildings()
+addFloorplan()
+removeFloorplan()
+addPOI()
+removePOI()
earns
has
reported on
defines
links to
scans
generates
defines
manages
creates
starts
creates
earns and spends
uses
displays
follows
contains
contains
has
links to